

INES ALONSO

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Research interests: interpretable, verifiable AI for quantum and physical sciences, with environmental and societal impact

EDUCATION

Florida Polytechnic University | GPA: 3.95/4.0

B.S. Computer Science, Minor in Applied Mathematics

Lakeland, FL, USA

Graduating December 2026

University of Barcelona

Physics undergraduate coursework (transferred to Florida Polytechnic)

Barcelona, Spain

Transferred June 2024

Relevant coursework: Machine Learning, Quantum Information & Computing, Numerical Methods, Multivariable Calculus.

Honors: SPIE@ICFO María Yzuel Fellowship; FPU Merit Scholarship; Esperanza Inc. Scholarship; President's List (3 terms)

RESEARCH EXPERIENCE

Research Intern in Computational Biophysics

Barcelona, Spain

Institute of Photonic Sciences (ICFO), SPIE@ICFO María Yzuel Fellowship

May 2025 – Present

Lewenstein (Quantum Optics Theory) & Liguori (Photon Harvesting) groups: how plants switch light harvesting to photoprotection

- Built a five-layer VAMPnet (PyTorch) to characterize PsbS/LHCII conformational kinetics across three pH conditions, learning nonlinear slow coordinates from 175 coarse-grained MD trajectories totaling **4,500 μ s** of sampling.
- Implemented out-of-core time-lagged independent component analysis (TICA), streaming covariance estimates over a **750,000 $\times$ 26,796** feature matrix (80 GB) to keep full residue-level resolution without downsampling or feature pruning.
- Built a reproducible Python framework (MDTraj, deeptime, scikit-learn) applying one Markov State Model validation battery (Chapman–Kolmogorov tests, spectral-gap analysis, held-out cross-replica scoring) across all three conditions.

Quantum Software Research Assistant

Lakeland, FL

Florida Polytechnic University · PI: Dr. Bradford Barker

February 2024 – Present

- Co-developed **q-sc-SF-EOM**, a quantum spin-flip EOM algorithm (Qiskit, PySCF) resolving ground and excited singlets in one diagonalization from a high-spin reference, cutting H_2 circuits from **52 to 16 CNOTs** and the H_4 excitation manifold **9 $\times$** .
- Derived a completeness framework quantifying the fraction of the $M_S = 0$ determinant space each operator manifold spans (100% at (2e,2o), 25% for H_4 , 6.7% for H_2O), predicting spin contamination and error *a priori* across five molecules.

PUBLICATIONS & PRESENTATIONS

- I. Alonso** et al., “The q-sc-SF-EOM quantum algorithm for ground- and excited-states in molecules and solids,” *manuscript in preparation* for arXiv submission, September 2026.
- Poster** on q-sc-SF-EOM, IEEE Quantum Week (QCE26), Toronto, Canada **Sept 2026**
- Poster** (selected, Undergraduate), STEM Research Competition, SHPE National Convention, Indianapolis, IN **Oct 2026**
- Oral and poster presentation**, NanoFlorida 2026 Conference, Lakeland, FL **Oct 2026**

SELECTED RESEARCH PROJECTS

Agentic MSM Benchmark Engine: Multi-Agent AI for Physics Discovery | *Independent Research*

July 2026

- Architected a multi-agent LLM system (Orchestrator/Optimizer/Validator) that designs analysis pipelines for noisy simulations; a deterministic physics gate, not the LLM, accepts results (4/4 runs converged; 20 physics rejections).
- Built an MSM pipeline recovering transition rates within **1.87%** of the exact escape-rate law (error budget $\approx$ **3.16%**).
- Formally verified the pipeline's ground-truth physics constants in Lean 4 (Mathlib) using an automated theorem prover.

Extending Hierarchical Reasoning Models with Attention Feedback | *Python, PyTorch*

November 2025

- Extended the Hierarchical Reasoning Model with bidirectional cross-attention feedback for two-timescale multi-hop reasoning.
- Outperformed Llama 3.1 and Kimi-K2 on HotPotQA (**85.15% F1**) with a **60.6M-parameter** model, **<1%** of their size.

NASA Florida Space Grant Consortium | *Space Trek Program, Center for Space Education*

March 2025

- Designed an altitude-triggered parachute deployment system for a hydrogen balloon (Arduino, BMP280, MOSFET).
- Programmed real-time magnetic-field, cosmic-ray, and solar-UV data acquisition across altitudes (Geiger, UV, IMU sensors).

LEADERSHIP

Sustainability President | *Florida Polytechnic Student Government Association*

May 2025 – July 2026

- Founded the university's Sustainability Agency and led 8 ambassadors; built a 100 m² campus garden on a \$2,500 budget and ran 200+ attendee events with Residence Life and sponsors.

Vice President of Internal Affairs | *SHPE (2025 Certified Chapter Leader)*

January 2025 – July 2026

- Ran 10 professional-development workshops and 8 heritage events; liaised with the Provost's Office and Career Services.

TECHNICAL SKILLS

Programming: Python, C/C++, SQL, Lean 4, LaTeX, Git/GitHub, Linux/Ubuntu, Docker, REST APIs, Pydantic

ML & Scientific: PyTorch, scikit-learn, deeptime, PyEMMA, MDTraj, NumPy/SciPy, Qiskit, PySCF, CUDA, MATLAB, LLM APIs

Training: IBM Qiskit Global Summer School 2025 (Quantum Excellence Badge); Barcelona Supercomputing Center CUDA Course

Languages: Spanish (native), Catalan (native), English (fluent)